

Kimura Kenjiro Research Project: Background, Purpose, and How We'll Proceed

Purpose of this document: to give the team a shared baseline on why this project is starting, what we're going to understand, and what we want to discuss next.

Part 0: For the Reader

This project does not yet have a fixed strategy. We are in the phase of **getting the facts straight together, then deciding strategy**. This document is not the answer; it's the common ground for the discussion.

What is decided, and what is not, up front:

Decided

- The subject of investigation is Professor Kimura Kenjiro at Kobe University (and his subsidiary, Integral Geometry Science, hereafter "IGS")
- The team's near-term tasks are (a) understanding the professor's theory and (b) gathering material to make a strategic choice
- We will not run from the math or the papers

Not decided

- How we relate to Kimura / IGS (joint research, adjacent territory at a distance, implementing without partnering, etc. — multiple options remain open)
 - Which application line to focus on (medical / battery / security / infrastructure / other)
 - How deep into the math we go (intuitive grasp / can follow the equations in the papers / can implement)
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Part 1: Background and Purpose

1.1 Why we are starting this project (the meta-hypothesis)

The hypothesis we're operating on can be condensed into one sentence:

In academia, there are many researchers doing work with significant business value who are not fully exploiting it on the business side. Small teams like ours have room to step in there.

If this hypothesis holds, our strategy is not to bet on a single researcher. It is to **systematically discover these "academically strong, commercially under-exploited" players, and then**

choose the best counterpart and territory to act on. The Kimura case is our first concrete instance.

1.2 Why we start with Kimura Kenjiro

He has heavy media exposure, abundant papers, and many video resources, which means **the outline of his theory is unusually accessible from outside**. That makes him an ideal first case study. The technology itself is high-quality, the application range is wide, and if there is room for us to enter, we'll learn a lot. Conversely, if we conclude there is no room, that is also a valuable hypothesis test.

1.3 Who is Kimura Kenjiro (the 5-minute version)

- Professor at the Center for Mathematical and Data Sciences (CMDs), Kobe University
- CEO of Integral Geometry Science (IGS)
- In 2012, derived the analytical solution to the "inverse problem of wave scattering," long held to be one of the unsolved problems in the history of applied mathematics
- The same year, founded Integral Geometry Instruments LLC, the predecessor of IGS
- Background: Kyoto University (quantum physics, atomic force microscopy) → moved to Kobe University in 2008 and pivoted to imaging methodology
- Awards and partnerships: 1st Japan Medical Research and Development Grand Prize (President's Award, 2017), AMED medical-engineering collaboration, JAXA Solar System Frontier project lead (2022), NEDO Deep Tech Startup Support Program lead (2024), investment from TOPPAN Holdings, extensive media coverage (NHK, TV Tokyo, NewsPicks, Hori-emon programs, etc.)

1.4 What he solved (the 3-minute version)

He solved the **inverse problem of wave scattering**, analytically.

- **Forward problem:** given the shape and position of an object, calculate **how radio waves, sound waves, or microwaves scatter** when applied to it. This is solvable from the basic equations of classical physics.
- **Inverse problem:** from observed scattering alone, **reconstruct the unknown object's internal structure**. This was the long-standing hard problem in applied mathematics.

Conventionally, the inverse problem was approached by numerical methods: posit an initial model, run forward simulations repeatedly until they fit the observed data. The computational cost is enormous, and convergence is not guaranteed if the initial model is far off.

What Kimura achieved is **a theoretical framework that yields a closed-form analytical solution to this inverse problem**. Specifically, he established what is called **multi-static scattering field theory** (or multi-path scattering field theory). In this framework, observation points, illumination points, and scattering points are treated as free coordinate variables in a higher-dimensional space, the partial differential equation governing the scattering field admits an

analytical solution, and finally the higher-dimensional coordinates are converged onto the actual 3-D space to reconstruct the object's interior.

If you grasp this one paragraph, Part 1's job is done. Part 2 is about how we go from here down to the actual equations.

1.5 Full inventory of application lines

The mathematical core is single, but it branches into multiple application lines depending on the type of wave and the use case. This matrix is the most important map for this project.

Application line	Wave	Use case	Commercialization status	Owner
Microwave mammography	Microwave	Breast cancer screening	In clinical trial; JIS standard already obtained	IGS
Inline storage battery inspection	Magnetic field	Non-destructive imaging of current-density distribution inside Li-ion batteries	NEDO-supported; targeting mass production by 2026	IGS
Super security gate	Microwave	Walk-through detection of concealed dangerous items	Prototype / proof-of-concept	IGS
FOCUS series (electronic component failure analysis)	Magnetic field	Visualization of current paths inside semiconductors and electronic devices	Already deployed at multiple companies	IGS
Infrastructure non-destructive inspection	Electromagnetic	Concrete cracks, rebar fracture and corrosion	Co-developed with major inspection-equipment manufacturers; tunnel inspection from a moving train demonstrated	IGS + majors
Road subsidence early detection	Electromagnetic	Underground voids and degradation	Mentioned in media; product status undisclosed	IGS
MOON scattering field tomographic imaging	Electromagnetic	Tomographic imaging of the lunar deep structure	JAXA research project	JAXA collaboration
Ultrasonic imaging	Ultrasound	High-resolution 3-D imaging via 2-D arrays	Research stage	Kobe Univ.

Application line	Wave	Use case	Commercialization status	Owner
"Lost face" of the Tower of the Sun	Electromagnetic	Internal investigation of structures	NHK collaboration completed	NHK collaboration
Brain visualization / brain repair	Unspecified	Conceptual stage	Not yet started	Kobe Univ.
Full elucidation of chemical-reaction principles	Unspecified	Conceptual stage	Not yet started	Kobe Univ.
Oil and geological exploration	Electromagnetic / elastic	Detection of underground fluids	Research stage	Kobe Univ.

This matrix is the starting point when we later discuss "where, if anywhere, is there room for us to enter."

1.6 What's already happening: the state of IGS

As context for whatever we decide to do, **the counterpart is already quite far along.**

- **Founded:** 2012 (including the predecessor LLC)
- **Funding:** January 2024, Series C 1st close: ~JPY 2.0 B third-party allotment (SBI Investment) plus ~JPY 2.5 B NEDO Deep Tech grant, totaling ~JPY 4.5 B. PitchBook's cumulative figure: ~USD 41 M
- **Shareholders:** SBI Investment, TOPPAN Holdings (via CVC), Asahi Kasei, Japan Asia Investment, etc.
- **Track record on software revenue:** the FOCUS series has been on sale since around 2014; Kobe University's press release at the time cited revenue exceeding JPY 200 million
- **Patents:** the core patent "Scattering tomography method and scattering tomography device" is granted in 9 countries including JP, US, CN, EU. **More than 100 patents** in total across IGS-related filings
- **Media coverage:** NHK, TV Tokyo, NewsPicks, HORIE ONE, Nature Portfolio, etc.

1.7 What we are trying to do (current state)

To repeat: strategy is not yet fixed. Listing the directions currently on the table:

1. **Partner with IGS** (joint research, consulting, licensing in)

2. **Take application domains that IGS is not pursuing** (work in adjacent territory, while watching the patent walls)
3. **Build the upper layer on top of IGS's core technology** (AI image reading, PACS integration, SaaS dashboards, etc.)
4. **Set Kimura aside and partner with a different researcher** (focus on the meta-hypothesis itself; find another under-exploited player)
5. **Withdraw** (after due diligence, conclude there is no path to win here)

We don't yet have the material to choose. The next section lays out what to gather.

1.8 Near-term team tasks

We are starting with theory understanding. The shared near-term task is for every team member to understand the professor's theory at least well enough to explain it in their own words; designated members go further. Part 2 covers the how.

The other tasks needed to support the eventual strategic choice (comprehensive patent mapping, competitor / adjacent-player map, codifying the meta-hypothesis screening criteria) are organized as key questions in Part 3. We'll take them up in turn, in the discussion, as the theory-understanding work progresses.

Part 2: How We Approach the Math

2.1 Setting the goal (this is the discussion)

For this project, **how deep we go into the math is itself coupled with the strategic choice.** This is one of the central things we want to discuss using this document.

Goal level	Content	Rough timeline	Required talent
L1: conceptual grasp	Can explain forward vs. inverse problem, analytical vs. numerical solution, and the outline of multi-static scattering field theory in their own words	2-4 weeks	Whole team
L2: can follow the equations	Can reproduce on a whiteboard the basic equations of multi-static scattering field theory, the boundary conditions, and the derivation of the analytical solution	3-6 months	MSc or above in PDEs / integral geometry / inverse problems; ideally PhD
L3: can implement	Reproduces toy cases numerically; gains intuition for resolution, SNR, and computational cost	6-12 months	L2 talent + someone with numerical-implementation experience (FDTD, etc.)
L4: production-grade	Can debug with real instrument data including calibration; approaches IGS's published benchmarks	1.5-3 years	L3 team + hardware front-end (antenna / transducer / A-D) designer

Open questions:

- Is our goal L1, L2, L3, or L4?
- If L4: **the instrument know-how does not appear in public materials**, so we cannot reasonably reproduce it without partnering with IGS. This strongly constrains the strategy.
- If L2: viable strategies are "partner," "build the upper layer," or "focus on the meta-hypothesis."
- If L1: most likely we wind down the Kimura case and consolidate around the meta-hypothesis.

Setting this goal is something **the team should discuss collectively** while looking at the table above. Bring an opinion to the meeting.

2.2 Learning roadmap: four layers of public resources

We've sorted available resources into four layers by difficulty. Whichever goal (L1-L4) we choose, the lower layers must be passed through.

Layer 1: Overview (accessible to anyone)

Everyone should start here. To grasp that the theory exists and to get its outline.

Medium	Title	URL	Time
Video	"Trying to understand 'inverse problem of wave scattering' even by 1 mm" — Hashimoto Kouji's Science Newsletter (TV Tokyo BIZ)	https://www.youtube.com/watch?v=SgjF7qFB4r8	20-30 min
Video	HORIE ONE × Kimura Kenjiro: "Now possible to see through oil fields, cancer cells, and even what's in your bag"	https://www.youtube.com/watch?v=KI-dS88OF58	60 min
Video	TV Tokyo BIZ Breakthrough: "Math genius Professor Kimura Kenjiro"	https://www.youtube.com/watch?v=AYr4zh3X6Dg	30 min
Video	Even Hori-emon got excited! Seeing through the invisible world with the power of math (HORIE ONE, separate episode)	https://www.youtube.com/watch?v=k4eWrcaN7Ks	30 min
Video	"Whither this country? 2" #86: Developing a groundbreaking breast-cancer screening device	https://www.youtube.com/watch?v=2NEaqxxJPvg	30 min
Article	Kobe University: "The science of looking beneath the surface" (English)	https://www.kobe-u.ac.jp/en/news/article/2019_06_17_01/	15 min
Article	Nikkei xTECH: "The equation that 'sees through' objects: toward practical use in breast cancer, battery defects, and weapon detection"	https://xtech.nikkei.com/atcl/nxt/column/18/00001/08617/	10 min

Medium	Title	URL	Time
Article	J-POWER: "Looking ahead with 'the science of seeing through' to the new shape of society" (mentions the "worm-eaten theory")	https://www.jpowers.co.jp/ge/82/opinion/index_01.html	10 min

Layer 2: Building mathematical intuition (undergraduate STEM level)

Mandatory for anyone going beyond L2. Equations appear, but a non-specialist can follow with effort.

Medium	Title	URL
Lecture video	Foundations of Computational Life Science 8: "An equation that saves lives," part 1 — Kimura Kenjiro	https://www.youtube.com/watch?v=3MpdiqcwEEE
Lecture video	Kobe University CMDS, frontier mathematics lecture (introduced via CMDS)	https://www.youtube.com/watch?v=SgjF7qFB4r8
Invited paper	Kimura, Kimura, "Multi-static Inverse Wave Scattering Theory and Microwave Mammography," Systems / Control / Information, 64(3), 2020	journal
Article	Kimura, "Analytical solution of the inverse problem of wave scattering and the realization of microwave mammography," Gakushikai Journal, 2022	Gakushikai
Conference paper	Kimura et al., "Foundations and applications of multi-path scattering field	https://www.jstage.jst.go.jp/article/jsapmeeting/2021.2/0/2021.2_50/_pdf/-char/ja

Medium	Title	URL
	theory," JSAP Autumn 2021 meeting	

Layer 3: The mathematical core (peer-reviewed papers and proceedings)

Required for L3 or above.

Medium	Title / source	URL
Proceedings	Kenjiro Kimura, "Inverse Scattering Field Theory," RIMS Kokyuroku No. 2186, Chap. 10, Kyoto University, 2021	https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/2186-10.pdf (note: scanned PDF)
Peer-reviewed	Kimura, Mima, Kimura, "Local electric current reconstruction theory for nondestructive inspection inside battery cell using magnetic field measurement," Subsurface Imaging Science & Technology, 1(1), 2017	journal
Peer-reviewed	Suzuki et al., "Non-destructive visualization of short circuits in lithium-ion batteries by a magnetic field imaging system," Japanese Journal of Applied Physics, 60(5), 056502, 2021	journal
Peer-reviewed	Kimura et al., "Development of Microwave Scattered Field Tomographic Imaging System and Clinical Trial Results," 26th Annual Meeting of the Japanese Breast Cancer Society, 2018	journal
Book chapter	Kimura et al., "Inverse Scattering Problem and Microwave Mammography," in <i>Screening and Risk Reduction Strategies for Breast Cancer</i> , pp. 163-175, 2023	book
Journal	Kimura et al., "Development of multistatic scattering field theory and actualization of microwave mammography," JSMI Report, 15(2), pp. 17-24, 2022	journal

Starting points for the publication list: **researchmap** (<https://researchmap.jp/read0140340>) and **J-GLOBAL** (https://jglobal.jst.go.jp/detail?JGLOBAL_ID=200901001633721127).

Layer 4: Patents (the scope of implementation rights)

- Core patent "Scattering tomography method and scattering tomography device" granted in 9 countries (JP, US, CN, EU, etc.)
- Over 100 patents filed across IGS-related applications
- Confirmed example: US20240280361A1 "Imaging device and imaging method" (published 2024, assigned to IGS, inventors: Kimura Kenjiro / Kimura Fumitoshi / Kimura Noriaki)

Starting points for cross-search:

- Google Patents: `inventor:"Kenjiro Kimura" + assignee:"Integral Geometry Science"`
- J-PlatPat (Japan Patent Office): search by inventor

The full patent mapping will be carved out as a separate task (see Part 3).

2.3 Pitfalls to anticipate

Three points where engineers new to this field tend to stumble.

(a) The asymmetry between forward and inverse problems Forward: "given an object, compute the scattering." Inverse: "given the scattering, infer the object." **They use the same physics, but they have very different mathematical character.** The inverse problem is generally ill-posed: existence, uniqueness, or stability with respect to data fails. Kimura's answer to this is: extend the scattering-field function into a higher-dimensional space, write a basic equation, and obtain the analytical solution.

(b) Analytical vs. numerical solutions Conventional approaches tried to crack the inverse problem by iterative numerical methods. No convergence guarantee, enormous compute cost. **An analytical solution means the answer can in principle be written in closed form.** That carries an overwhelming advantage in computational efficiency and in principled correctness. Note however that "an analytical solution exists" and "we can stably reconstruct from real instrument data" are different things — the former is mathematics, the latter is numerical analysis, signal processing, and calibration.

(c) Coordinate extension into higher-dimensional space Intuitively it's not obvious why scattering happening in 3-D space would be lifted into a higher-dimensional space. The reason is **to treat observation, illumination, and scattering points as independent coordinate variables**, which is what allows the scattering-field function to be written in PDE form. The final step — taking the higher-dimensional coordinates to converge onto the 3-D target region — is what corresponds to "imaging." **Numerical stabilization of this limit operation is most likely the implementation's crux**, and presumably where the patent claims are densely filed.

Part 3: Key questions to resolve going forward

After reading this document, the key questions we will need to work through. We split them into **questions that decide strategic direction** and **investigative tasks needed to support that decision**.

Questions that decide strategic direction

1. **On the meta-hypothesis:** do we keep "there are many under-exploited academic players with significant business value" as the project's central hypothesis?
2. **Position of the Kimura case:** do we treat it as the first instance for verifying the above hypothesis, or as a standalone Kimura project?
3. **Which application line to focus on:** from the matrix in §1.5, where do we go deep first?
4. **Goal for math understanding:** which of L1-L4?
5. **Withdrawal conditions:** spell out, in advance, what we'd have to learn for us to walk away.

Investigative tasks needed to support that decision

We'll take these up in turn, in the discussion, as the theory-understanding work progresses.

6. **Comprehensive patent mapping:** extract every patent assigned to or invented by IGS / Kimura and classify the claims. **This determines whether the "implement without partnering" scenario is feasible.**
7. **Competitor and adjacent-player map:** major players worldwide in microwave imaging, ultrasound inverse problems, magnetic-field imaging, etc.
8. **Codify the meta-hypothesis screening criteria:** write down the criteria for "under-exploited academic players" and re-evaluate Kimura against them.

Appendix: Glossary

- **Inverse problem of wave scattering:** from observed scattered waves, reconstruct the internal structure of the object that caused the scattering. Long held to be one of the unsolved problems in the history of applied mathematics
- **Multi-static scattering field theory** (multi-path scattering field theory): the framework Kimura established. Treats the observation, illumination, and scattering points as independent coordinate variables in a higher-dimensional space; obtains an analytical solution to the scattering-field equation
- **"Worm-eaten theory" (mushikui riron):** the popular-audience nickname Kimura uses for the multi-static scattering field theory
- **Forward problem:** from cause to effect (object → scattering)
- **Inverse problem:** from effect to cause (scattering → object)

- **Ill-posed problem:** a problem for which existence, uniqueness, or continuous dependence on data fails. Inverse problems are typically ill-posed
 - **Scattering tomography:** a method to tomographically image the interior of an object from scattered waves. The subject of Kimura's core patent
 - **Microwave mammography:** a breast-cancer screening device using microwaves. Works on dense breast tissue, where X-ray mammography struggles
 - **IGS:** Integral Geometry Science Inc., the company Kimura leads as CEO
 - **FOCUS series:** a non-destructive current-path imaging system for electronic components, on sale since the predecessor company
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Revision history

- v0.1 (initial Japanese version): draft of background, purpose, and math-learning roadmap
- v0.2: video links expanded to full URLs, two additional overview videos
- v0.3 (English translation): translated and adapted from v0.2
- v0.4: editorial cleanup
- v0.5: removed §2.4 and §2.5; repositioned Part 3 as "Key questions to resolve going forward"
- v0.6: §1.8 closing line now states "we are starting with theory understanding"
- v0.7: §1.8 narrowed to theory understanding; patent mapping and other tasks moved to Part 3, which is now split into "direction" questions and "investigative tasks"
- v0.8: softened "explain from memory" to "explain in their own words"